

Artificial Intelligence

BSc Computer Science - Durham University

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1 Module Introduction

Artificial Intelligence has the module code COMP2261, and is led by Dr Yang Long.

1.1 Assessment

Title	Opening	Due	Feedback	Professor	Weight
Machine Learning	9th Oct	15th Dec	21st Jan	Dr Yang Long	50%
AI Search	15th Jan	18th Feb	18th Mar	Prof Iain Stewart	25%
Bias in AI	5th Feb	30th Mar	6th May	Dr Shauna Concannon	25%

Table 1: Assessments for Artificial Intelligence.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	14:00 - 16:00	Computer Class	TLC025	-
Tuesday	16:00 - 17:00	Lecture	W103	Dr Yang Long
Friday	15:00 - 16:00	Lecture	W103	Dr Yang Long

Table 2: Michaelmas Term Timetable for Artificial Intelligence.

2 Machine Learning

This section is taught by Dr Yang Long. The recommended reading for this topic is:

- Mathematics for Machine Learning (Deisenroth, M.P., Faisal, A.A., and Ong, C.S., 2020).
- Mastering Machine Learning with scikit-learn (Hackeling, G., 2017).

2.1 Introduction to Machine Learning

2.2 Machine Learning Workflow and Coding Environment

2.3 Data Understanding

2.4 Data Preparation Workflow

3 Artificial Intelligence Search

This section is taught by Professor Iain Stewart. There is no recommended reading for this topic however there may be an additional video to watch after lectures to enable more content to be covered.

Watch the FAQs video.

3.1 Coding Real-World Problems for Search Tree Expansion

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

3.2 Uninformed Search: BFS, DFS, and Iterative Deepening

Watch ahead of time - this will not be covered in lectures. Week 5 covers no content.

- 3.3 General Heuristic Search: Greedy Best-First Search and A* Search**
- 3.4 A* Search and Deriving Iterative Deepening A* Search**
- 3.5 Heuristic Functions and Their Development**
- 3.6 Local Search: Hill-Climbing, Simulated Annealing, and Genetic Algorithms**
- 3.7 Swarm Behaviour, How Ants Forage, and a Simple S-ACO algorithm**
- 3.8 Applying Different Ant Systems to the TSP**
- 3.9 General Versions of Particle Swarm Optimisation and Differential Evolution**
- 3.10 Applying Particle Swarm Optimization to the TSP**

4 Ethics and Bias in AI

This section is taught by Dr Shauna Concannon. The recommended reading is handled on a weekly basis for this topic.

- 4.1 Introduction to AI Ethics and Related Concepts**
- 4.2 Societal Impacts of AI**
- 4.3 Defining Bias and Fairness in ML**
- 4.4 How to Build Better Systems**
- 4.5 Value Sensitive Design**
- 4.6 Evaluations and Audits**
- 4.7 Bias Mitigation Strategies**
- 4.8 Transparency and Accountability for Trustworthy Systems**
- 4.9 Explainable AI**
- 4.10 Future Trends and Ethical Frontiers**

Part I

Appendix

5 Bibliography

5.1 Module Professors

Module Lead: Doctor Yang Long

- Focus: Scalable AI solutions for large-scale healthcare applications.
- Email: yang.long@durham.ac.uk
- Office Hours: Monday 15:30 - 16:30 in Term 1.

Lecturer: Professor Iain Stewart

- Focus: Computational complexity, graph/group theory, and theoretical aspects artificial intelligence.
- Email: i.a.stewart@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00 in Term 2.

Lecturer: Doctor Shauna Concannon

- Focus: Computational linguistics, social interaction, and human-computer interaction.
- Email: shauna.j.concannon@durham.ac.uk
- Office Hours: Thursday 14:30 - 15:30 in Term 2.

5.2 References

5.3 List of Figures

5.4 Reading List

5.4.1 Machine Learning

Deisenroth, M.P., Faisal, A.A. and Ong, C.S. (2020). Mathematics for Machine Learning. [online] Cambridge University Press. Available at: <https://mml-book.com/> [Accessed 3 Oct. 2026].

Hackeling, G. (2017). Mastering Machine Learning with scikit-learn. 2nd edn. Packt Publishing.

5.4.2 Ethics and Bias in AI

Various, will be added over time.